



RKDF University

Value-Added courses imparting transferable and life skills

Course Name: SMART MANUFACTURING THROUGH DIGITAL MANUFACTURING

About The Course:

The future trend of manufacturing will predominantly include cyber-physical systems, the industrial internet of things and cloud technologies. Such a trend, generally termed as Industry 4.0 in Germany and Smart Manufacturing in North America, is expected to introduce new business models in manufacturing, new processes, new technology and finally new skills in the factory worker to cope with all of these changes.

Needless to say, this global trend has already taken root in India. Over the next decade, rapid advancements in Smart Manufacturing are expected in India, centered around empowering factory workers with new technology, adequate automation and assisted devices to perform work more effectively, efficiently and without errors. This would necessitate the development of the requisite skills and retraining of the current work force and simultaneous development of new curriculum within the academia to prepare students for this emerging manufacturing landscape.

Beyond providing an overview of the emerging concepts in manufacturing, this course will cover the engineering processes, practices, technology and applications of Smart Manufacturing. The objective of the course is to provide a strong orientation to both the industry and the students on the new advancements in manufacturing in general and the relevant features of Smart Manufacturing to an Indian context, in particular.

COURSE OBJECTIVES

By the end of the course, students will be able to

- To learn the fundamental principles in Machine to Machine (M2M) Communication
- To provide a hands-on introduction to design of mechatronic systems, namely sensors, actuators, interfaces, computer hardware, and control software, and enable understanding of the theory and practice of mechatronic systems integration.
- To analyze manufacturing systems in terms of material flow and storage, information flow, capacities, and times and durations of events.
- Understand basics of open source hardware/software, Mechaphonics, and mobile/web apps

COURSE OUTCOMES

At the end of this course, the students are expected to

- Understand the standards, protocols, and algorithms in M2M communication
- Implement the M2M Communication protocols in a prototype
- Understand the basic concepts of the main sensors used in electromechanical systems
- Understand the fundamental concepts of mechanical power transmission components, and pneumatic and hydraulic actuators
- The students will be able to understand the probability, queuing models, optimization, process analysis, and linear and dynamic systems.

COURSE CONTENTS

Introduction to smart manufacturing:

What is “smart manufacturing” really and how does it differ from conventional/legacy manufacturing-Smart Manufacturing. Processes- Three Dimensions: (1) Demand Driven and Integrated Supply Chains ;(2) Dynamically Optimized Manufacturing Enterprises (plant + enterprise operations);(3) Real Time, Sustainable Resource Management (intelligent energy demand management, production energy optimization and reduction of GHG)

Smart design/fabrication:

Smart Design/Fabrication - Digital Tools, Product Representation and Exchange Technologies and Standards, Agile (Additive) Manufacturing Systems and Standards. Mass Customization, Smart Machine Tools, Robotics and Automation (perception, manipulation, mobility, autonomy), Smart Perception – Sensor networks and Devices.

Smart Communication systems:

Information, Mobility, Communication Technologies, Protocols, Cyber Physical Systems – the next generation of Embedded Systems and Networks, IT and OT convergence, co-creation and collaboration enablement. Smart Cloud- Hyper scale Computing; Application Delivery Platforms and Platform as a Service; Intelligent Analytics Services.

Smart Applications:

Online Predictive Modeling, Monitoring and Intelligent Control of Machining/Manufacturing and Logistics/Supply Chain Processes; Smart Energy Management of manufacturing processes and facilities

Smart and Empowered Workers:

Eliminating Errors and Omissions, Deskilling Operations, Improving Speed/Agility, Improving Information Capture/Traceability, Improving Intelligent Decision Making under uncertainty Assisted/Augmented Production, Assisted/Augmented Assembly, Assisted/Augmented Quality, Assisted/Augmented Maintenance, Assisted/Augmented Warehouse Operations and Assisted Training